It may seem surprising that the central scheme is *more* accurate for a PDE which we never intended to approximate. However, considering the nature of the difference approximation, and that of the error when expressed in terms of a Taylor series expansion, this is actually not unusual. In fact, every difference approximation is more accurate for some equation, other than the one we intend to approximate, if some of the terms remaining in the Taylor series are included in the equation. The modified equation can help to characterize the behavior of the scheme. Here we see that the right-hand side of (6.6) is given by negative diffusion, a term that also appears when solving the heat equation backwards in time, cf. Example 2.1. Recall that this leads to exponential growth in the solution. It is thus not entirely surprising that the central difference approximation is only stable is the sense of $r \leq 1 + C\tau$, which permits exponential growth (cf. Remark 4.1).

6.2 The Upwind Scheme

The nature of the advection equation suggests that information comes only from "one side". We could try to respect this by using a one-sided stencil in the approximation of the spatial derivative. This leads to the *upwind scheme* for (5.10),

$$\frac{u_i^{n+1} - u_i^n}{\tau} + a\frac{u_i^n - u_{i-1}^n}{h} = 0, \qquad a > 0, \tag{6.8}$$

and

$$\frac{u_i^{n+1} - u_i^n}{\tau} + a\frac{u_{i+1}^n - u_i^n}{h} = 0, \qquad a < 0. \tag{6.9}$$

We can combine (6.8) and (6.9) into a scheme that works for both positive and negative values of a:

$$\frac{u_i^{n+1} - u_i^n}{\tau} + a\frac{u_{i+1}^n - u_{i-1}^n}{2h} - \frac{|a|}{2}\frac{u_{i-1}^n - 2u_i^n + u_{i+1}^n}{h} = 0 \tag{6.10}$$

It is easily seen that (6.10) reduces to (6.8) and (6.9) for positive a, and negative a, respectively. We formally define the upwind scheme for the periodic problem:

Scheme 6 (Upwind Scheme) For $i = 1, \ldots N-1$ and $n = 0, 1, \ldots M-1$ we set

$$\begin{aligned} u_i^{n+1} &= u_i^n - \frac{\nu}{2}(u_{i+1}^n - u_{i-1}^n) + \frac{|\nu|}{2}\left(u_{i-1}^n - 2u_i^n + u_{i+1}^n\right), \\ u_i^0 &= u_0(ih). \\ u_i &= u_{i+N} \end{aligned} \tag{6.11}$$

where $\nu = a\tau/h$. Here h and τ are the grid parameters corresponding to the grids $\mathcal{G}_h$ (cf. (4.4)), and $\mathcal{G}_\tau$ (cf. (4.5)), respectively.

To show consistency of the scheme, assume u is a smooth solution of the advection equation, and consider

$$r_h(x,t) := \frac{u(x,t+\tau) - u(x,t)}{\tau} + a\frac{u(x+h,t) - u(x-h,t)}{2h} - \frac{|a|}{2}\frac{u(x-h,t) - 2u(x,t) + u(x+h,t)}{h} \tag{6.12}$$

The third term is similar to the second order consistent approximation of u_{xx} first introduced in eq. (3.2). Comparing to (3.2), we easily deduce

$$\frac{|a|}{2}\frac{u(x-h,t) - 2u(x,t) + u(x+h,t)}{h} = \frac{|a|h}{2}u_{xx}(x,t) + \mathcal{O}(h^3) \tag{6.13}$$

The first two terms in (6.12) can be expanded in a Taylor Series as in (6.1), and (6.2), respectively. Combining all the terms, we have

$$r_h = \underbrace{u_t + au_x}_{=0} + \frac{1}{2}\tau u_{tt} - \frac{|a|h}{2}u_{xx} + \mathcal{O}(\tau^2 + h^2).$$

Therefore the upwind scheme is seen to be consistent of (only) first order in space and time. Note, however:

Theorem 6.2 *The Upwind Scheme 6 is consistent of second order in space and time with the equation*

$$u_t + au_x = \frac{|a|h}{2}\left(1 - \frac{\tau|a|}{h}\right)u_{xx}. \tag{6.14}$$

Proof Let u be smooth and solve (6.14). Substitute u into the Upwind Scheme, and define $r_h(x,t)$ as in (6.12). We obtain

$$r_h(x,t) = u_t(x,t) + au_x(x,t) + \frac{1}{2}\tau u_{tt}(x,t) - \frac{|a|h}{2}u_{xx}(x,t) + \mathcal{O}(\tau^2 + h^2).$$

Note that we cannot cancel the first two terms, if we assume u solves (6.14). However, we can write

$$\begin{aligned} u_t &= -au_x + \mathcal{O}(h) \\ \Rightarrow u_{tt} &= -au_{xt} + \mathcal{O}(h) \\ &= -au_{tx} + \mathcal{O}(h) \\ &= a^2 u_{xx} + \mathcal{O}(h) \end{aligned}$$

So we can write

$$\begin{aligned} r_h &= \underbrace{u_t + au_x - \frac{|a|h}{2}(1 - |\nu|)\,u_{xx}}_{=0} + \mathcal{O}(\tau^2 + h^2 + \tau h) \\ &= \mathcal{O}(\tau^2 + h^2) \end{aligned}$$

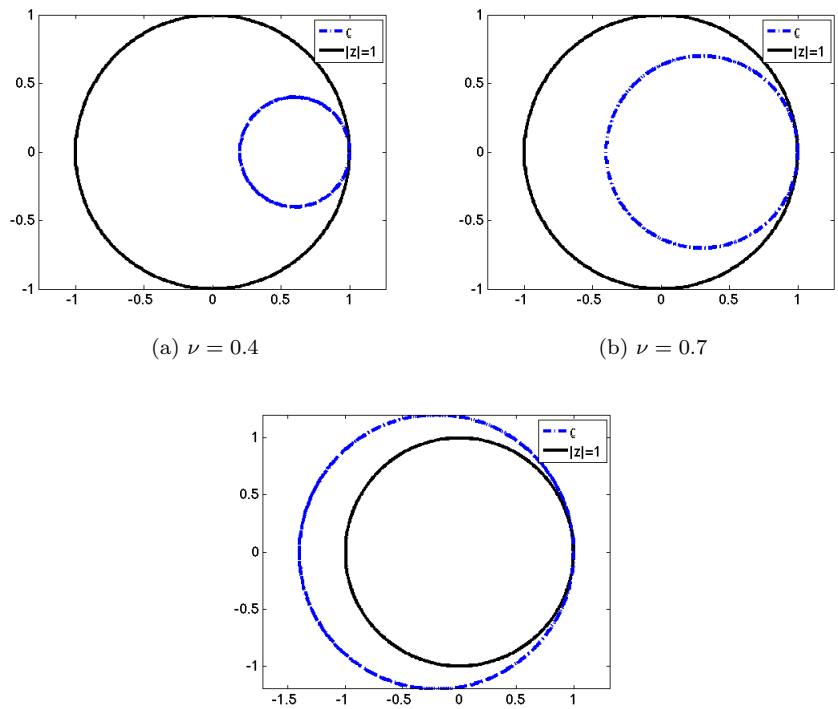
where the definition of $\nu = a\tau/h$ has been used. (Note: Young's inequality $ab \le 0.5(a^2 + b^2)$ has also been used.) ∎

The right-hand side of (6.14) has two terms. The second term comes from the central difference approximation to the first spatial derivative. It is the same term that appears in the modified equation of the central scheme (cf. eq. (6.6)). The first term comes from the extra second-difference term that appears in the upwind scheme (6.10). We would thus expect the Upwind scheme to be stable as long as $|\nu| \le 1$: In this case the right-hand side of the modified equation stays non-negative, and we avoid the negative diffusion which we obtained for the central scheme. Loosely speaking, one can say that the job of the extra second order difference operator appearing in the upwind scheme (6.10) is to balance the negative diffusion term that comes from the central approximation.

For the stability analysis we use the von Neumann test. We first assume $a > 0$. The amplification factor reads

$$|\rho_k(\tau,h)| = \left|1 - \nu + \nu e^{-\hat{\imath}kh}\right|$$

For constant ν, the loci of the amplification factor lie on a circle of radius ν. The circle is centered around $1-\nu$ on the real axis. For $0 < \nu \leq 1$, this circle lies within the unit circle. (For $\nu = 1$ the two circles coincide.) For $\nu > 1$ the loci of the amplification factor fall outside the unit circle, and the scheme is unstable. (For $a < 0$, we get a similar statement, in that $-1 \leq \nu < 0$ for stability.) Figure 6.1 illustrates the situation for $a > 0$.



(a) $\nu = 0.4$

(b) $\nu = 0.7$

(c) $\nu = 1.2$

Figure 6.1: Upwind scheme for the advection equation. Loci of the amplification factor.

6.3 The Lax-Friedrichs Scheme

Another simple way to cure the central scheme is to replace u_i^n in (6.3) with an average:

$$\frac{1}{\tau}\left(u_i^{n+1} - \frac{1}{2}(u_{i+1}^n + u_{i-1}^n)\right) + a\frac{u_{i+1}^n - u_{i-1}^n}{2h} = 0.$$

This can be written

$$\frac{u_i^{n+1} - u_i^n}{\tau} + a\frac{u_{i+1}^n - u_{i-1}^n}{2h} - \frac{1}{2\tau}(u_{i+1}^n - 2u_i^n + u_{i-1}^n) = 0.$$

This is the *Lax-Friedrich Scheme*. We formally define the scheme for the periodic problem:

Scheme 7 (Lax-Friedrich Scheme) For $i = 1, \ldots N-1$ and $n = 0, 1, \ldots M-1$ we set

$$\begin{aligned} u_i^{n+1} &= u_i^n - \frac{\nu}{2}\left(u_{i+1}^n - u_{i-1}^n\right) + \frac{1}{2}(u_{i+1}^n - 2u_i^n + u_{i-1}^n), \qquad n = 0, 1, 2 \ldots \\ u_i^0 &= u_0(ih). \\ u_i &= u_{i+N} \end{aligned} \tag{6.15}$$

where $\nu = \frac{\tau a}{h}$. Here h and τ are the grid parameters corresponding to the grids $\mathcal{G}_h$ (cf. (4.4)), and $\mathcal{G}_\tau$ (cf. (4.5)), respectively.

Compared to the central scheme, the effect of the modification is thus to add a term that is consistent with the second derivative, i.e.

$$\frac{u(x-h,t) - 2u(x,t) + u(x+h,t)}{2\tau} = \frac{h^2}{2\tau} u_{xx}(x,t) + \mathcal{O}(\frac{h^4}{\tau}),$$

for u smooth enough. Noting again (6.1), and (6.2), we can write the consistency error of the Lax-Friedrich scheme

$$r_h = u_t + au_x + \frac{\tau}{2} u_{tt} + \frac{\tau^2}{6} u_{ttt} - a\frac{h^2}{3} u_{xxx} - \frac{h^2}{2\tau} u_{xx} + \mathcal{O}(\tau^3 + h^4 + \frac{h^4}{\tau}).$$

We can see that, if $\tau = \mathcal{O}(h)$, the scheme is first order consistent in time and space[2].

Without proof we state a modified equation for the Lax-Friedrich scheme. One can show that the scheme is second order consistent with the equation

$$u_t + au_x = \frac{h^2}{2\tau}\left(1 - \nu^2\right) u_{xx}, \tag{6.16}$$

which again suggests that the scheme is stable as soon as $|\nu| \leq 1$. Indeed the von Neumann test yields

$$\begin{aligned} |\rho_k| &= \left| 1 + \hat{\imath}\nu \sin(kh) + \frac{1}{2}\left(e^{\hat{\imath}kh} - 2 + e^{-\hat{\imath}kh}\right) \right| \\ &= \left| 1 + \hat{\imath}\nu \sin(kh) + \frac{1}{2}\left(2\cos(kh) - 2\right) \right| \\ &= |\hat{\imath}\nu \sin(kh) + \cos kh| \\ &= \sqrt{\nu^2 \sin^2(kh) + \cos^2(kh)} \end{aligned}$$

For given ν, the loci of the amplification factor thus lie on an ellipse with semi axes $|\nu|$ and 1. For $|\nu| < 1$, the ellipse lies inside the unit circle, while for $|\nu| = 1$ the ellipse becomes the unit circle. For $|\nu| > 1$ the values of the amplification factor lie outside the unit circle. Figure 6.2 shows a few examples.

6.4 The Lax-Wendroff Scheme

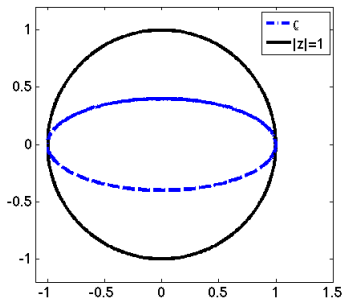
We are inconvenienced by the fact that the upwind scheme and the Lax-Friedrich scheme are both only first order consistent. We would like to have a scheme, which is consistent of higher order. To this end, we try to incorporate the consistency analysis already when we design the scheme. Consider the following Taylor series:

$$u(x, t+\tau) = u(x,t) + \tau u_t(x,t) + \frac{\tau^2}{2} u_{tt}(x,t) + \frac{\tau^3}{6} u_{ttt}(x,t) + \mathcal{O}(\tau^4).$$

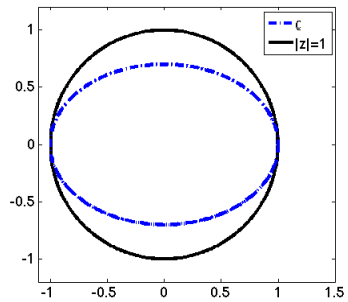
If u satisfies the advection equation, we can replace

$$\begin{aligned} u_t &= -au_x \\ u_{tt} &= -au_{xt} = -au_{tx} = a^2 u_{xx}, \end{aligned}$$

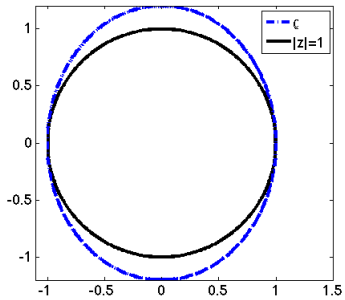
[2]Note that if we take $\tau = \mathcal{O}(h^2)$, the scheme is not consistent with the advection equation anymore! This has to do with the way the diffusion term is scaled. The factor ν does not appear there, and if $\tau = \mathcal{O}(h^2)$, then the diffusion term that is scaled with h^2/τ in the consistency error does not vanish as the mesh is refined.



(a) $\nu = 0.4$



(b) $\nu = 0.7$



(c) $\nu = 1.2$

Figure 6.2: Lax-Friedrich Scheme for the advection equation: Loci of the amplification factor.

which leads to

$$u(x,t+\tau) = u(x,t) - a\tau u_x(x,t) + \frac{a^2\tau^2}{2}u_{xx}(x,t) + \mathcal{O}(\tau^3),$$

and hence

$$\frac{u(x,t+\tau)-u(x,t)}{\tau} = -au_x(x,t) + \frac{a^2\tau}{2}u_{xx}(x,t) + \mathcal{O}(\tau^2). \tag{6.17}$$

When adding a discretization for the space derivative in the advection equation, we must make sure that the first two terms on the right-hand side of eq. (6.17) are cancelled, and all consistency errors remain of second order. The first term will be cancelled (with consistency error of second order), if we use a central discretization for u_x (cf. eq. 6.2). The second term can be cancelled by noting

$$\frac{a^2\tau}{2}\frac{u(x-h,t)-2u(x,t)+u(x+h,t)}{h^2} = \frac{a^2\tau}{2}u_{xx}(x,t) + \mathcal{O}(\tau h^2), \tag{6.18}$$

Define the discretization as

$$\frac{u_i^{n+1}-u_i^n}{\tau} + a\frac{u_{i+1}^n - u_{i-1}^n}{2h} - \frac{a^2\tau}{2}\frac{u_{i-1}^n - 2u_i^n + u_{i+1}^n}{h^2} = 0.$$

Or, multiplying the equation with τ, we have

Scheme 8 (Lax-Wendroff Scheme) For $i = 1,\dots N-1$ and $n = 0,1,\dots M-1$ we set

$$\begin{aligned} u_i^{n+1} &= u_i^n - \frac{\nu}{2}\left(u_{i+1}^n - u_{i-1}^n\right) + \frac{\nu^2}{2}(u_{i+1}^n - 2u_i^n + u_{i-1}^n), \\ u_i^0 &= u_0(ih). \\ u_i &= u_{i+N} \end{aligned}$$

where $\nu = \frac{\tau a}{h}$. Here h and τ are the grid parameters corresponding to the grids $\mathcal{G}_h$ (cf. (4.4)), and $\mathcal{G}_\tau$ (cf. (4.5)), respectively.

For the consistency analysis we simply add the consistency errors of (6.17), (6.2), and (6.18). We have

$$r_h = \mathcal{O}(\tau^2 + h^2) \tag{6.19}$$

Hence, the scheme is consistent of second order for $\tau = \mathcal{O}(h)$.

We begin the stability analysis by noting that the Lax-Wendroff scheme does not meet the positivity condition established in Section 4.3. In fact, if we write the scheme as

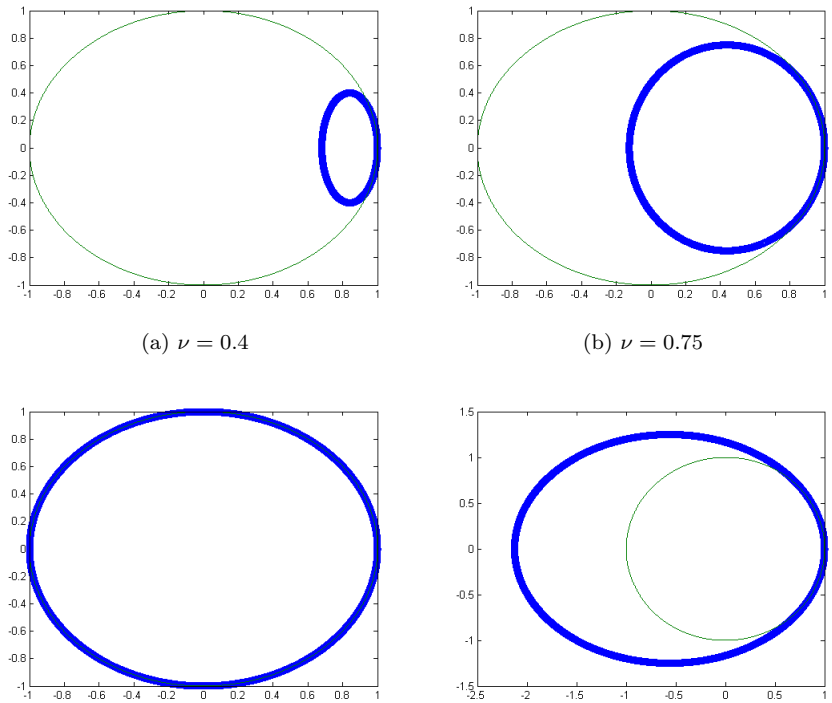
$$u_i^{n+1} = u_i^n - \frac{\nu}{2}(u_{i+1}^n - u_{i-1}^n) + \frac{\nu^2}{2}(u_{i-1}^n - 2u_i^n + u_{i+1}^n) \tag{6.20}$$

$$= \frac{\nu}{2}(\nu+1)u_{i-1}^n + (1-\nu^2)u_i^n + \frac{\nu}{2}(\nu-1)u_{i+1}^n \tag{6.21}$$

we see that, if $\nu > 1$, the second coefficient is negative, while for $\nu < 1$ the last coefficient is negative[3]. To show that we *do* meet the von-Neumann stability condition, we compute the amplification factor

$$\begin{aligned} |\rho_k| &= \left|\frac{1}{2}\nu(\nu+1)e^{-\hat{\imath}kh} + 1 - \nu^2 + \frac{1}{2}\nu(\nu-1)e^{\hat{\imath}kh}\right| \\ &= \left|\nu^2\cos(\xi) + 1 - \nu^2 - \hat{\imath}\nu\sin(\xi)\right| \\ &= \left|1 - \nu^2(1-\cos(\xi)) - \hat{\imath}\nu\sin(\xi)\right| \end{aligned}$$

[3] If $\nu = 1$ the scheme is reduced to the exact solution $u_i^{n+1} = \nu u_{i-1}^n$. This is a special artifact of the advection equation which has such a simple solution structure.



(a) $\nu = 0.4$ (b) $\nu = 0.75$

(c) $\nu = 1$ (d) $\nu = 1.25$

Figure 6.3: Graphical illustration of the loci of the amplification factor for different values of ν. The unit circle is included for reference.

Take the magnitude to obtain

$$\begin{aligned}|\rho_k|^2 &= \left(1-\nu^2(1-\cos(\xi))\right)^2 + \nu^2\sin^2(\xi)\\ &= 1 - 2\nu^2(1-\cos(\xi)) + \nu^4(1-\cos(\xi))^2 + \nu^2\sin^2(\xi)\\ &= 1 - \nu^2\left\{2 - 2\cos(\xi) - \nu^2(1-\cos(\xi))^2 - \sin^2(\xi)\right\}\end{aligned}$$

Using $1-\sin^2(\xi) = \cos^2(\xi)$, we get

$$\begin{aligned}|\rho_k|^2 &= 1 - \nu^2\left\{1 - 2\cos(\xi) + \cos^2(\xi) - \nu^2(1-\cos(\xi))^2\right\}\\ &= 1 - \nu^2\left\{(1-\cos(\xi))^2 - \nu^2(1-\cos(\xi))^2\right\}\\ &= 1 - \nu^2(1-\nu^2)(1-\cos(\xi))^2\end{aligned}$$

All factors in the second term are non-negative if $|\nu| \le 1$, which means that $|r| \le 1$. On the other hand, for $\nu > 1$ the factor $1-\nu^2$ becomes negative, and $|r| > 1$. Figure 6.3 shows a graphical illustration of the loci of the amplification factor in the complex plane.

6.5 Summary of Discretization for the Advection Equation

It is interesting to note that the schemes we have defined in this chapter form a family of schemes which can be written in the form

$$u_i^{n+1} = u_i^n - \frac{\nu}{2}\left(u_{i+1}^n - u_{i-1}^n\right) + \frac{s}{2}(u_{i+1}^n - 2u_i^n + u_{i-1}^n), \qquad n = 0, 1, 2\ldots$$

with

- $s = 1$: Lax-Friedrich Scheme
- $s = |\nu|$: Upwind Scheme
- $s = \nu^2$: Lax-Wendroff Scheme
- $s = 0$: Central Scheme

The only difference between the schemes is apparently the scaling of the diffusion coefficient. However, this can make quite a big difference. We have already seen that the Central Scheme, which has the least diffusion (i.e. negative diffusion), is not stable. Apparently *some* extra diffusion is needed. The Lax-Friedrich scheme has the most diffusion. The Upwind scheme has less diffusion in the stable regime, and the Lax-Wendroff scheme has even less. In fact, the Lax-Wendroff scheme is also of second order consistent, because it cancels *exactly* the leading-order negative diffusion term of the central scheme. So it would appear that in devising the Lax-Wendroff scheme, we have found just the right amount of diffusion to keep the scheme stable (at least under von Neumann test), and at the same time improve the accuracy to second order. But there is also a slight problem with the Lax-Wendroff scheme in that we do not have positive coefficients. We will further investigate the situation in the next section.

6.6 Dissipation and Dispersion

We analyze the behavior of the error for the Lax-Friedrichs scheme and the Lax-Wendroff scheme. We shall see that these are entirely different.

First we consider a general description of a traveling wave,

$$e^{\hat{\imath}(kx-\omega t)} = e^{\hat{\imath}k(x-ct)}, \tag{6.22}$$

where ω is the angular frequency, k is the wavenumber, and $c = \omega/k$ is the phase speed. We say there is "no dispersion" if c is constant, meaning the phase velocity does not depend on the wavelength of the traveling wave. This is the same as saying that ω is a linear function of k:

$$\omega(k) = ck \qquad \Rightarrow c = \frac{\omega}{k} = \frac{d\omega}{dk} \tag{6.23}$$

On the other hand, we say there is "dispersion", if $\omega(k)$ is a nonlinear function of k, i.e. the phase velocity is not a constant, and depends on the wavenumber. In this case we have

$$c_g = \frac{d\omega}{dk} \neq \frac{\omega}{k} = c, \tag{6.24}$$

where c_g is usually called the group velocity.

The relevance is that we know how to produce traveling wave solution using the advection equation. Consider the periodic problem

$$\begin{aligned} u_t + au_x &= 0, \qquad x \in [0, 2\pi), t > 0 \\ u(x,0) &= \phi(x), \\ u(x+2\pi,t) &= u(x,t), \qquad \text{for all } x \\ \phi(x+2\pi,t) &= \phi(x,t), \qquad \text{for all } x \end{aligned}$$

and let

$$\phi(x) = e^{\hat{\imath}kx}.$$

Then the solution is given by

$$u(x,t) = e^{\hat{\imath}k(x-at)}.$$

The solution is a traveling wave with phase speed $c = a$. It should be pointed out that the advection equation requires that there be no dispersion: Every wave travels at the same speed,

regardless of its wavelength. The traveling wave solution has significance beyond this simple case. The reason is that we can think of more general initial conditions as being composed of waves, by way of Fourier series [4]:

$$\phi(x) = \sum_{k=-\infty}^{\infty} c_k e^{\hat{\imath} kx}$$

The coefficients can be obtained as

$$\frac{1}{2\pi}\int_0^{2\pi} e^{-\hat{\imath} mx}\phi(x)\,dx = \frac{1}{2\pi}\sum_{k=0}^{\infty} c_k \int_0^{2\pi} e^{\hat{\imath}(k-m)x}\,dx$$

where we assume that we can exchange order of integration and differentiation. Using

$$\frac{1}{2\pi}\int_0^{2\pi} e^{\hat{\imath}(k-m)x}\,dx = \delta_{m,k},$$

each coefficient is determined by

$$c_k = \frac{1}{2\pi}\int_0^{2\pi} e^{-\hat{\imath} kx}\phi(x)\,dx.$$

If we view the initial conditions as superposition of waves, the exact solution should be

$$u(x,t) = \sum_{k=-\infty}^{\infty} c_k e^{\hat{\imath} k(x-at)}. \tag{6.25}$$

Again, it should be noted that the advection equation has no dispersion. All the components of the wave packet travel with the same speed. The waves "stay" together.

6.6.1 The Lax-Friedrichs Scheme Re-Visited

We have noted in section 6.3 that the Lax-Friedrich scheme is second-order consistent with the *modified* equation

$$u_t + au_x = \frac{h^2}{2\tau}\left(1 - \left(\frac{\tau a}{h}\right)^2\right) u_{xx} \tag{6.26}$$

If we choose $\frac{\tau|a|}{h} \leq 1$ the scheme will be stable, and we can write the modified equation

$$u_t + au_x = \mu u_{xx}. \tag{6.27}$$

where $\mu > 0$. However, the solution to this equation does not propagate waves like the advection equation. We expect the solution to be close and try

$$u(x,t) = e^{\hat{\imath}(kx-\omega(k)t)}$$

Substituting

$$u(x,t) = e^{\hat{\imath}(kx-\omega(k)t)} \tag{6.28}$$

into the modified equation (6.27), one obtains

$$-\hat{\imath}\omega e^{\hat{\imath}(kx-\omega(k)t)} + \hat{\imath} ake^{\hat{\imath}(kx-\omega(k)t)} = \mu(\hat{\imath}k)^2 e^{\hat{\imath}(kx-\omega(k)t)}$$

Setting

$$\omega = ak - \hat{\imath}\mu k^2$$

[4] We will treat Fourier series only informally to illustrate a basic point. In particular we will not address the issue of convergence of Fourier series here.

we satisfy the modified equation (6.27). Substituting this into (6.28), we have

$$e^{\hat{\imath}(kx-(ak-\hat{\imath}\mu k^2)t)} = e^{\hat{\imath}k(x-at)}e^{-\mu k^2 t}$$

It can be seen that here is an exponential damping term, while the phase velocity shows no dispersion. (All waves still travel with speed a.) The damping term causes the excessive dissipation of the Lax-Friedrich scheme. (Recall, the scheme is only first order accurate for smooth solutions. In the exact solution to the advection equation, there is no decay of the solution at all!)

It should be noted that these characteristics are those of the exact solution of the modified equation. Since the Lax-Friedrich scheme is a better approximation to the modified equation, than it is for the advection equation, it inherits this behavior. This does not mean that the Lax-Friedrich scheme has no dispersion at all. These results merely indicate that the high-dissipation dominates the error of the Lax-Friedrichs scheme, and dispersive error is higher order small. (If we include more terms in the modified equation, we also see a smaller dispersion error.)

Also note that the Upwind scheme also has a modified equation with μu_{xx} on the right hand side (with a slightly different definition of μ). So everything said here about the Lax-Friedrichs scheme applies to the Upwind Scheme as well.

6.6.2 The Lax-Wendroff Equation Re-Visited

Without proof we state the modified equation for the Lax-Wendroff Scheme:

$$u_t + au_x = Du_{xxx}, \tag{6.29}$$

where $D < 0$ in the stable regime. The Lax-Wendroff scheme approximates this equation to one order higher that it does the linear advection equation. We again try to solve the modified equation for a traveling wave solution, and try

$$u(x,t) = e^{\hat{\imath}(kx-\omega(k)t)}.$$

Substituting this into (6.29) yields

$$-\hat{\imath}\omega(k)e^{\hat{\imath}(kx-\omega(k)t)} + \hat{\imath}ake^{\hat{\imath}(kx-\omega(k)t)} = D(\hat{\imath}k)^3 e^{\hat{\imath}(kx-\omega(k)t)}$$

The modified equation (6.29) is satisfied if

$$\omega(k) = ak + Dk^3.$$

The waves thus have a phase velocity of $c = a + Dk^2$, which is not constant! Hence, we have dispersion. While the solution will still be a combination of traveling waves, if initial conditions of the form (6.25) are given, the wave components all travel at different speeds. This becomes in particular relevant if the "frequency content"of the initial condition is high, meaning there is comparatively large magnitude of many coefficients c_k in (6.25). It is known from Fourier analysis that for smooth functions the coefficients c_k decay exponentially, so high frequencies carry negligible weight. For this reason, the dispersion error is not visible, or even significant. (Recall that the Lax-Wendroff scheme is second order accurate for smooth solutions!) For non-smooth solutions, however, the coefficients c_k decay much more slowly. In particular for functions that are not even continuous the coefficients c_k decay only as $1/k$. In particular, for such solutions the Lax-Wendroff scheme exhibits oscillations that are caused by the diverging waves.

We have thus seen that the error of the Lax-Friedrich scheme (and by the same token, that of the Upwind Scheme) is dominated by diffusion. This comes from the diffusive right-hand side in the modified equation. The error of the Lax-Wendroff scheme in contrast is dominated by dispersion. The question is, which is "better"? We have seen that the absence of excessive diffusion leads to higher order accuracy for the Lax-Wendroff scheme. So for smooth solutions, this scheme is clearly better. The main drawback is for nonlinear equations, which we have not discussed yet. We shall see that for nonlinear hyperbolic equations we must admit discontinuous solutions, or we have no solutions at all. (The precise mechanism by which this is accomplished will be detailed later.) The dispersive nature of the Lax-Wendroff scheme leads to oscillations near discontinuities, which in turn can destroy stability. We will deal with this shortly.